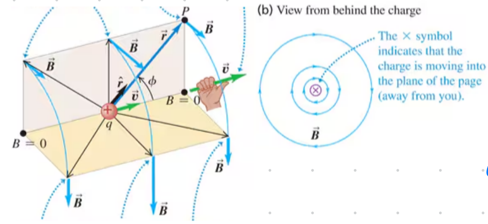
28.1 Magnetic field of Moving Charge

$\vec{B}$ is perpendicular to the plane containg $\vec{r}$ and $\vec{v}$.



$$B = \frac{\mu_0}{4\pi} \cdot \frac{|q| v \sin\phi}{r^2}$$

max B when $\phi = 90°$.

$$\boxed{\vec{B} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \hat{r}}{r^2}}$$

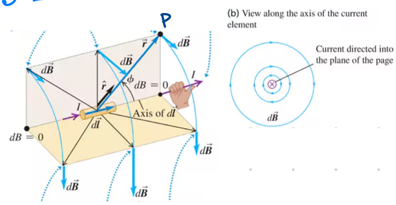
magnetic constant (μ_0); charge (q); velocity ($\vec{v}$); unit vector from point charge (source point) to field point ($\hat{r}$); distance from source (r^2)

mag. field due to point charge w/ constant velocity ($\vec{B}$)

28.2 Magnetic Field of a current Element

$$dQ = nqA\,dl$$

$\vec{B} \perp$ to $\vec{r}$ and $\vec{l}$



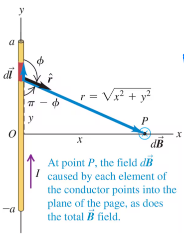
$$dB = \frac{\mu_0}{4\pi} \frac{I\,dl \sin\phi}{r^2}$$

$$\boxed{d\vec{B} = \frac{\mu_0}{4\pi} \frac{I\,d\vec{l} \times \hat{r}}{r^2}}$$

current (I); vector length of element in direction of current ($d\vec{l}$)

$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{I\,d\vec{l} \times \hat{r}}{r^2}$$

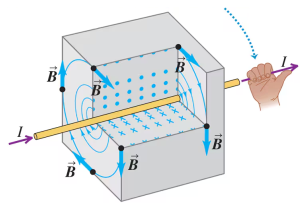
28.3 Magnetic field of straight current-carrying conductor



M-field near straight, current-carrying conductor:

$$\boxed{B = \frac{\mu_0 I}{2\pi r}}$$

current (I); distance from conductor (r)



surface current density:

$$\boxed{K = \sigma v}$$

charge / area (σ); velocity (v)

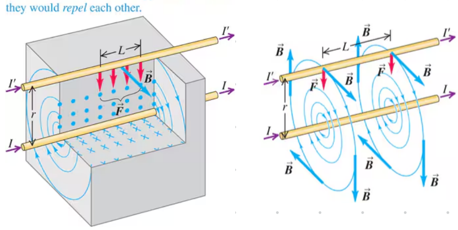
infinite moving sheet of charge:

$$\boxed{B = \frac{\mu_0 K}{2}}$$

28.4 Force between parallel conductors

The magnetic field of the lower wire exerts an attractive force on the upper wire. By the same token, the upper wire attracts the lower one.

If the wires had currents in *opposite* directions, they would *repel* each other.



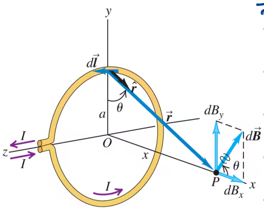
M-force per unit length between 2 long, straight, ∥ current-carrying conductors:

$$\boxed{\frac{F}{L} = \frac{\mu_0 I I'}{2\pi r}}$$

current in 1st conductor (I); current in 2nd conductor (I'); distance between conductors (r)

- ∥ conductors carrying current in the <u>same</u> direction <u>attract</u> each other.
- ∥ conductors carrying current in <u>opposite</u> directions <u>repel</u> each other.

28.5 Magnetic field of circular current loop



use RHR

$\vec{B}$ at P only has x-component:

$$B_x = \frac{\mu_0 I a}{4\pi (x^2+a^2)^{3/2}} \int dl \qquad \int dl = 2\pi a$$

M-field on axis of circular current-carrying loop $\Rightarrow$ $\boxed{B_x = \frac{\mu_0 I a^2}{2(x^2+a^2)^{3/2}}}$

current: I; a = radius of loop; x = distance along axis from center to field point

solenoid: $\boxed{B = \mu_0 n I}$ — n = # turns per unit length

For coil of N circular loops:

on axis of N loops: $B_x = \frac{\mu_0 N I a^2}{2(x^2+a^2)^{3/2}}$

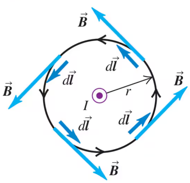
$= \frac{\mu_0 \mu}{2\pi(x^2+a^2)^{3/2}}$ → magnetic moment of N loops: $\mu_N = N I \pi a^2$

at center of N circular loops → $\boxed{B_x = \frac{\mu_0 N I}{2a}}$ — N = # of loops, I = current, a = radius

28.6 Ampere's law

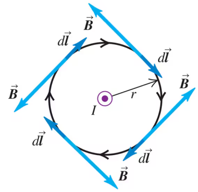
(a) Integration path is a circle centered on the conductor; integration goes around the circle counterclockwise.

Result: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$



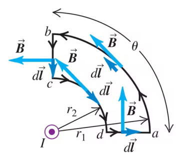
(b) Same integration path as in (a), but integration goes around the circle clockwise.

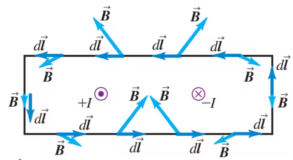
Result: $\oint \vec{B} \cdot d\vec{l} = -\mu_0 I$

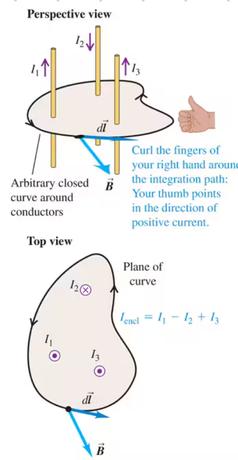


(c) An integration path that does not enclose the conductor

Result: $\oint \vec{B} \cdot d\vec{l} = 0$







Ampere's law: If we calculate the line integral of the magnetic field around a closed curve, the result equals μ_0 times the total enclosed current: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{encl}$.

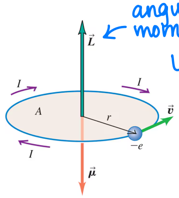
line integral around closed path

Ampere's law: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{encl}$

$\vec{B}$: M-field; $d\vec{l}$: vector segment of path; μ_0: M-constant; I_{encl}: net current enclosed by path

28.8 Magnetic materials

current of electron:



angular momentum: $L = mvr$

$$I = \frac{e}{T} = \frac{ev}{2\pi r}$$

e = charge; T = orbital period

$$\mu = \frac{evr}{2} = \frac{e}{2m} L$$

Curie's law:

$$M = C\frac{B}{T}$$

M = magnetization; C = constant; B → M-field; T → temp.

magnetization of a material:

$$\vec{M} = \frac{\vec{\mu}_{tot}}{V}$$ — V = volume

m-field from current + surrounding magnetized material

$$\vec{B} = \vec{B}_0 + \mu_0 \vec{M}$$

permeability of material

$$\mu = K_m \mu_0$$

K_m = relative permeability (>1)

magnetic susceptibility: $\chi_m = K_m - 1$